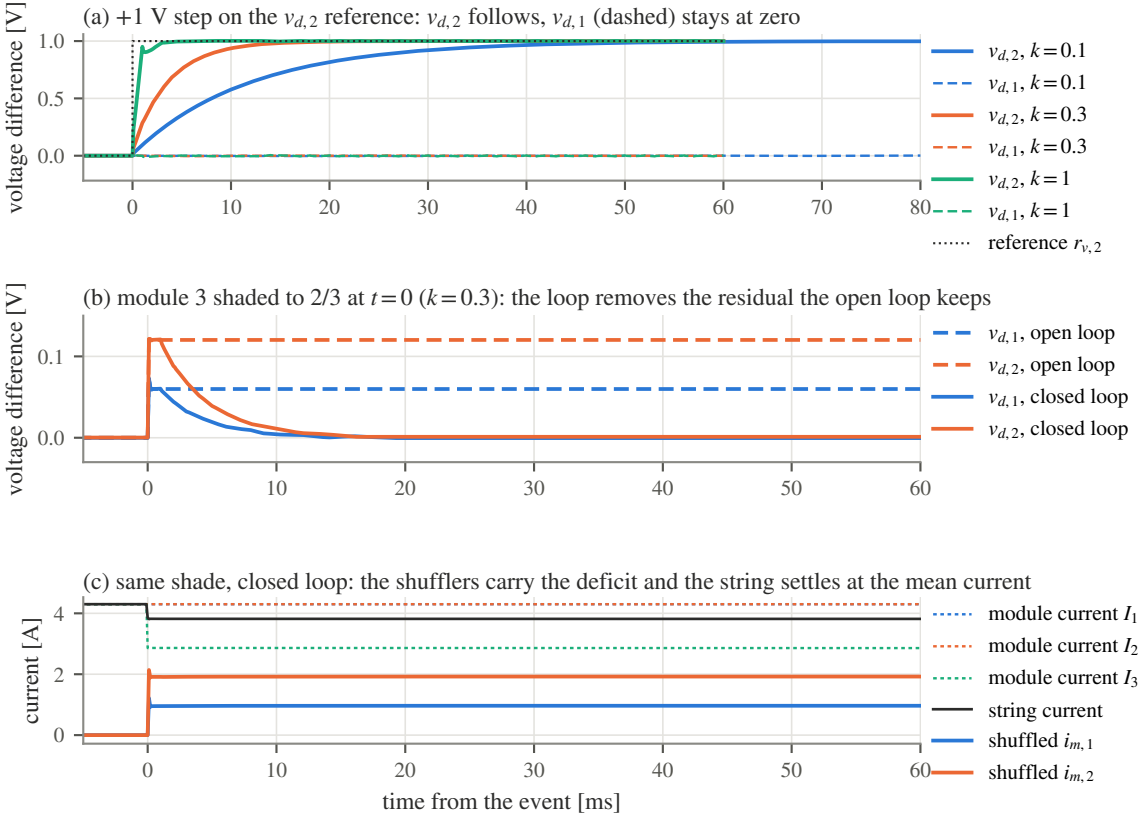
(a) +1 V step on the $v_{d,2}$ reference: $v_{d,2}$ follows, $v_{d,1}$ (dashed) stays at zero
voltage difference [V]
1.0
0.5
0.0
0
10
20
30
40
50
60
70
80
$v_{d,2}$, $k=0.1$
$v_{d,1}$, $k=0.1$
$v_{d,2}$, $k=0.3$
$v_{d,1}$, $k=0.3$
$v_{d,2}$, $k=1$
$v_{d,1}$, $k=1$
reference $r_{v,2}$
(b) module 3 shaded to 2/3 at $t=0$ ($k=0.3$): the loop removes the residual the open loop keeps
voltage difference [V]
0.1
0.0
0
10
20
30
40
50
60
$v_{d,1}$, open loop
$v_{d,2}$, open loop
$v_{d,1}$, closed loop
$v_{d,2}$, closed loop
(c) same shade, closed loop: the shufflers carry the deficit and the string settles at the mean current
current [A]
4
2
0
0
10
20
30
40
50
60
time from the event [ms]
module current I_1
module current I_2
module current I_3
string current
shuffled $i_{m,1}$
shuffled $i_{m,2}$